

Temporal and spatial evidence limits of trained PING

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Abstract

A frozen pyramidal-interneuron gamma (PING) network, whose trained weights remain fixed during evaluation, is tested under complementary temporal and spatial reductions of Modified National Institute of Standards and Technology (MNIST) digit evidence. It classifies continuously streamed digits without retraining, but a duration $\times$ input-rate sweep reveals a failure floor below 15 ms. At fixed 200 ms presentation and readout windows, performance remains at 10-class chance through 0.5 Hz and becomes clearly informative by 2 Hz. Separately, foreground pixels are permanently removed from binarized images and presented to both PING and a width-matched artificial neural network (ANN). PING is competitive at intermediate deletion but reaches chance by retention $q = 0.02$. Together, the curves delimit temporal, event-rate, and spatial evidence regimes for a future variable-rate training experiment.

Methods

Streaming duration and encoding rate

The trained baseline from the canonical PING experiment contains 1024 excitatory (E) and 256 inhibitory (I) cells. It was trained on one MNIST digit per 200 ms trial using random seeds 42, 43, 44. Each seed identifies an independent training run with a reproducible random initialization and data order. Everything here is **inference-only** at the trained timestep $\Delta t = 0.1$ ms; the weights are never updated. The two-dimensional sweep averages over all 3 seeds; the single-stream examples use seed 42.

A stream of digits, each shown for τ ms, is classified in one forward pass. The *only* change from training is a sliding readout window:

1. **Encode** each digit as a Poisson spike train over τ ms across 784 input channels, one per pixel. Each channel generates independent random spikes. The stated encoding rate is the expected number of spikes per second for a full-intensity pixel; lower pixel intensities reduce that rate proportionally.
2. **Concatenate** the per-digit trains into one input stream. At segment index k , the Poisson rate switches instantaneously at the boundary

$$t_k = k\tau. \quad (1)$$

Here t_k is the boundary time of segment k , k is the segment index, and τ is the duration of each segment.

3. **Run once** through the trained network at the trained Δt , without retraining.
4. **Integrate evidence** in a non-spiking output leaky integrator, one unit per class. A leaky integrator accumulates incoming spikes while gradually discounting older input:

$$v_{\text{out}}(t) = \beta_{\text{out}} v_{\text{out}}(t-1) + \frac{1 - \beta_{\text{out}}}{\Delta t} s^E(t-1) W_{\text{out}}. \quad (2)$$

Here t is the discrete timestep; $v_{\text{out}}(t)$ is the vector of output-unit states; β_{out} is their leak factor, the fraction of the previous output state retained for one timestep; Δt is the simulation timestep; $s^E(t-1)$ is the E-cell spike vector at the preceding timestep; and W_{out} is the trained E-to-output weight matrix.

5. **Read a sliding window.** Average v_{out} over the trailing τ -window and apply a softmax:

$$\text{logits}(t) = \frac{\Delta t}{\tau} \sum_{u=t-w+1}^t v_{\text{out}}(u). \quad (3)$$

$$p(\text{class}, t) = \text{softmax}(\text{logits}(t)). \quad (4)$$

Here $\text{logits}(t)$ is the class-evidence vector; u indexes timesteps in the window; τ is the presentation duration; w is its number of timesteps; and $p(\text{class}, t)$ is the softmax-normalized class-probability vector. Softmax converts the logits into non-negative class probabilities that sum to one. The readout-window duration is matched exactly to the current digit's presentation duration:

$$T_{\text{readout}} = T_{\text{presentation}} = \tau. \quad (5)$$

Here T_{readout} is the duration over which output evidence is averaged, $T_{\text{presentation}}$ is the time for which the digit is shown, and τ is that common duration.

The corresponding window length is

$$w = \frac{\tau}{\Delta t}. \quad (6)$$

Every digit is therefore read over exactly its own presentation duration; readout duration is not varied independently. At training the average ran over the *whole* trial; the trailing matched-duration window is the single change.

6. **Predict per segment** at the end of the digit's τ -window according to

$$\hat{c}(t) = \arg \max_c p(\text{class} = c, t). \quad (7)$$

Here c indexes the 10 digit classes, and $\arg \max$ selects the class with the largest probability.

The output leak is

$$\beta_{\text{out}} = \exp(-\Delta t / \tau_{\text{out}}). \quad (8)$$

Here τ_{out} is the output-unit time constant, which controls how quickly accumulated output evidence decays, and $\exp$ denotes the exponential function. The probability trace $p(\text{class}, t)$ is the network's online class confidence.

The grid uses presentation durations 10 ms, 15 ms, 25 ms, 40 ms, 50 ms, 75 ms, 100 ms, 200 ms and input rates 5 Hz, 10 Hz, 25 Hz, 50 Hz, 100 Hz, 200 Hz per channel. This gives 48 cells with 1200 classified segments per cell.

To resolve the encoding-rate floor below the grid, additional evaluations use rates 0.01 Hz, 0.025 Hz, 0.05 Hz, 0.1 Hz, 0.25 Hz, 0.5 Hz, 1 Hz, 2 Hz, 3 Hz while holding both presentation and readout at 200 ms. Each cell contains 10 streams of 10 digits for every trained seed. The 5 Hz, 10 Hz, 25 Hz, 50 Hz, 100 Hz, 200 Hz points use the same fixed-duration protocol and come from the corresponding grid row.

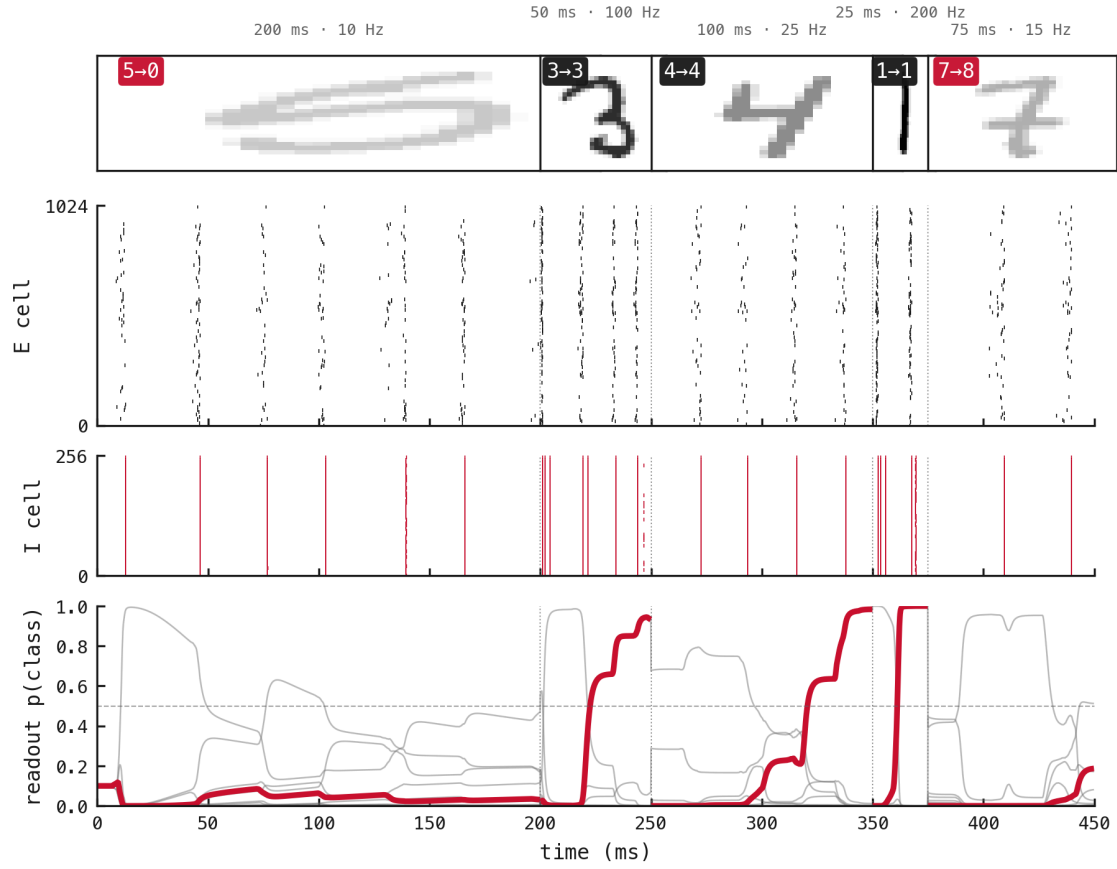
Foreground-retention calibration

The spatial protocol uses the same MNIST split and has two parts:

1. Train 3 seeds of a width-matched artificial neural network (ANN) with 784 inputs, one rectified-linear hidden layer of 1024 units, and 10 outputs. The hidden width matches the PING E population, not its recurrent E/I architecture. A rectified-linear unit outputs zero for a negative input and otherwise passes the input unchanged. Training uses 15 epochs, or complete passes through the training set, batches of 256 images per weight update, and learning rate 0.001, the step size of each update.
2. Binarize each held-out image, meaning an image excluded from training, at intensity 0: pixels above the threshold become unit-valued foreground and all others become zero-valued background. Retain every foreground pixel independently with probability q . Retention $q = 1$ leaves the foreground intact and $q = 0$ removes it. The ANN calibration uses 10 independent mask realizations per image. The matched comparison uses 100 fixed held-out examples and identical masks for every ANN and PING seed; PING encodes them at 25 Hz for 200 ms.

Results

Streaming classification and temporal evidence



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Figure 1: Classification when presentation duration and encoding rate vary between segments. The segment conditions are 200 ms at 10 Hz; 50 ms at 100 Hz; 100 ms at 25 Hz; 25 ms at 200 Hz; 75 ms at 15 Hz. Thumbnail opacity increases with encoding rate. The middle panels plot E- and I-cell spike rasters against time (ms); the lower panel plots class probability against time (ms), with the true class emphasized in red. The label-to-prediction pairs are 5→0, 3→3, 4→4, 1→1, 7→8, giving 3 of 5 correct segments.

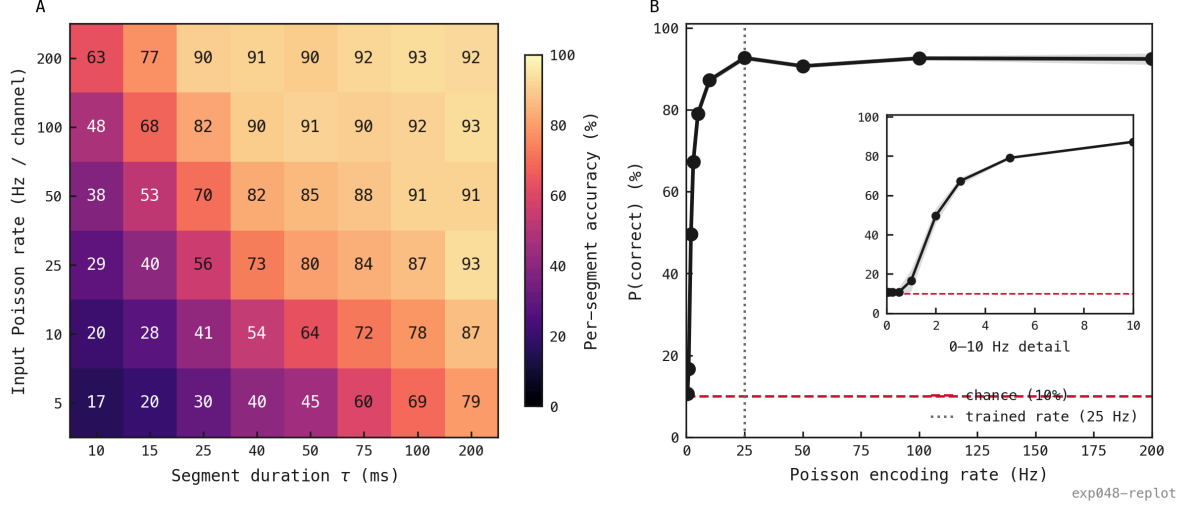


Figure 2: Temporal and encoding-rate limits of the frozen PING classifier. **(A)** Per-segment accuracy (%) is shown for presentation duration (ms, horizontal) and Poisson encoding rate (Hz per channel, vertical), using 1200 segments per cell. **(B)** Probability of a correct classification (%) is plotted against encoding rate (Hz) with presentation and readout fixed at 200 ms. The inset enlarges the linear 0.01–10 Hz interval without changing the axis scale. The dashed line marks 10-class chance and the dotted line the 25 Hz training rate. Accuracy stays at its empty-input floor, the accuracy obtained when almost no input spikes arrive, through 0.5 Hz, becomes informative by 2 Hz, and reaches 79.1% at 5 Hz.

The fixed-duration rate curve distinguishes a nonviable encoder regime from ordinary classification errors under weak evidence. In the variable-condition stream, the first failed segment received 10 Hz for 200 ms, yet that condition reaches 87.3% across the population. Its error is therefore natural trial-level variation, not evidence that 10 Hz is intrinsically too low. The other failed segment, presented at 15 Hz for 75 ms, is likewise above the empty-input rate floor, although its shorter window supplies less total evidence. Rates below 0.5 Hz are not useful operating points; 2 Hz is the lowest clearly informative tested rate and 5 Hz is a practical lower bound for future sweeps.

Spatial evidence calibration

The architecture-matched ANN remains above chance until foreground retention falls to $q = 0.005$, fewer than one visible foreground pixel per image on average.

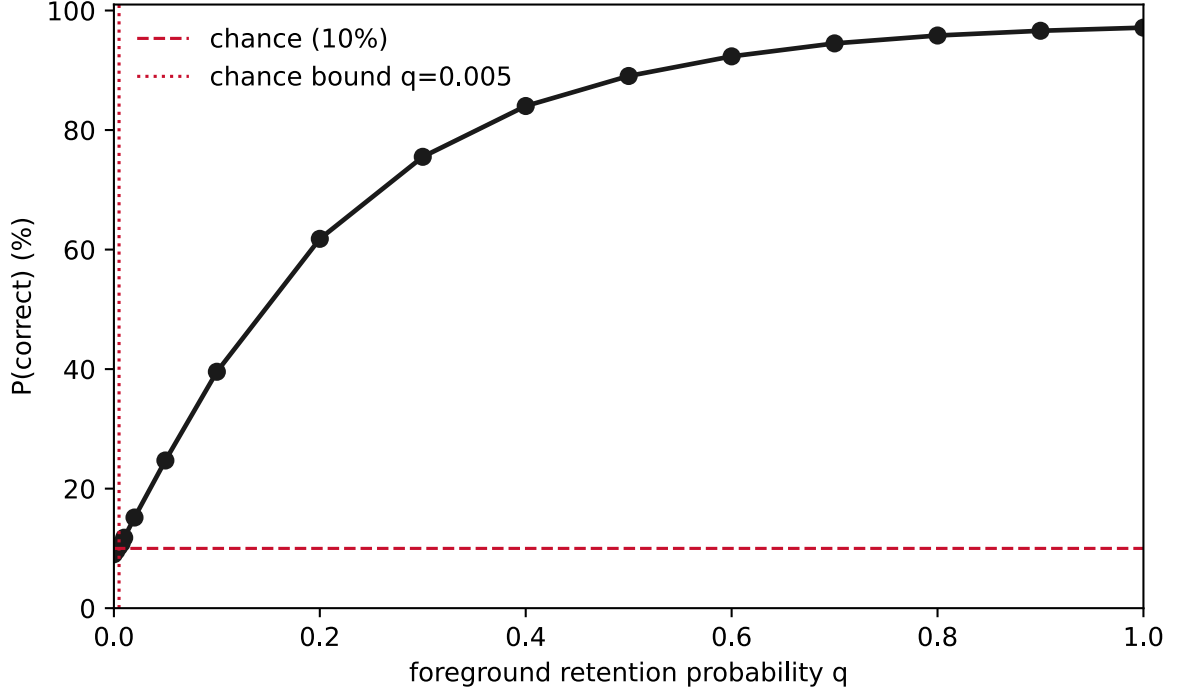


Figure 3: Held-out ANN accuracy as foreground evidence is removed. Probability of a correct classification (%) is plotted against foreground retention q , the independent probability that a foreground pixel remains visible. Points are means across 3 ANN seeds and the band is one standard error, the estimated uncertainty of that mean across seeds. The dashed line marks 10-class chance; the dotted line marks the measured chance-region bound at $q = 0.005$, the highest tested retention whose 95% confidence interval across seeds still contains chance accuracy.

Under identical masks, neither classifier is uniformly better. With 29.5 visible pixels on average ($q = 0.2$), PING reaches 68.7% against the ANN’s 60.7%. They coincide near 45% at $q = 0.1$. PING still leads at $q = 0.05$, then reaches chance at $q = 0.02$ while the ANN remains above it.

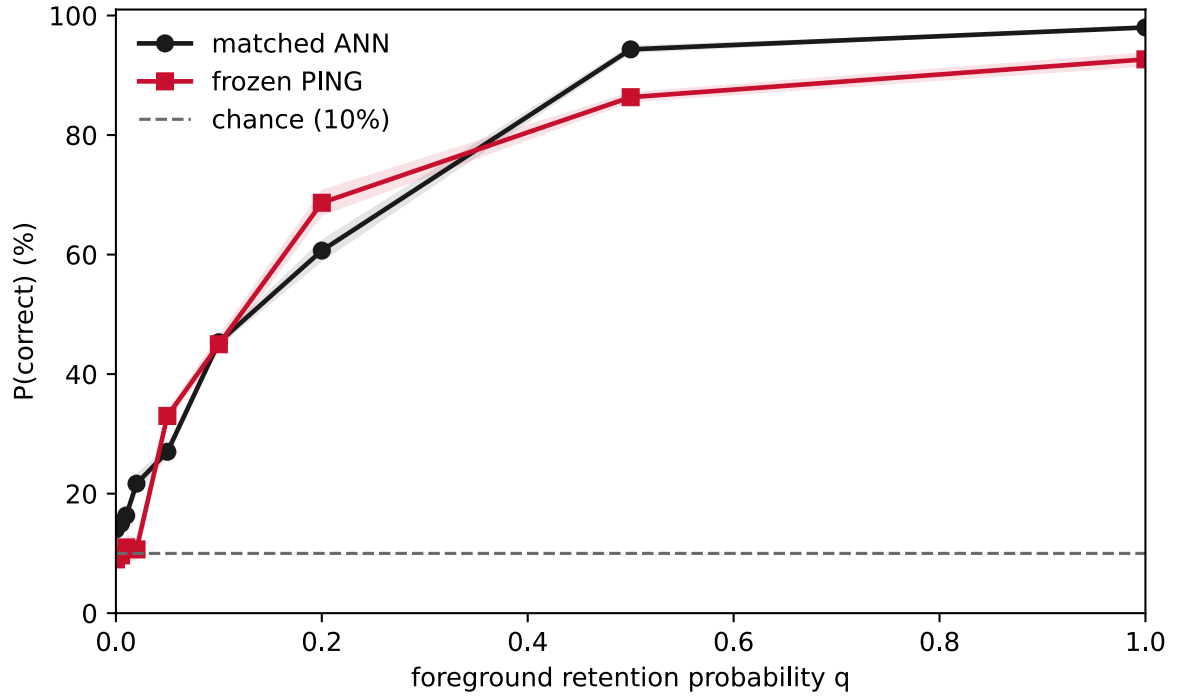


Figure 4: Width-matched ANN and frozen PING accuracy under identical spatial deletion. Probability of a correct classification (%) is plotted against foreground retention q . Black circles denote ANN and red squares PING; bands show one standard error across trained seeds. Each point uses 100 fixed held-out examples and identical independently sampled foreground masks. PING runs at 25 Hz for 200 ms. Neither classifier dominates across the full retention range.

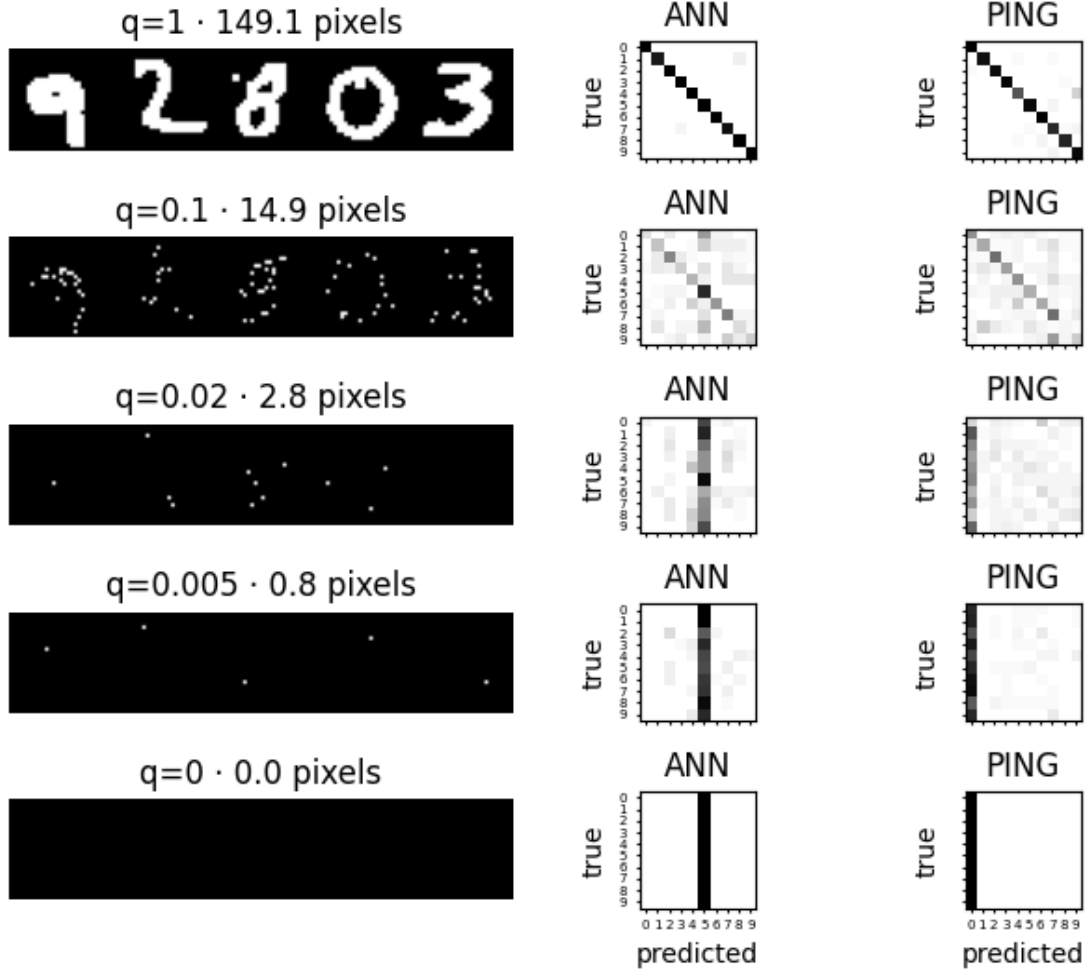


Figure 5: Stimulus and error structure across the matched masking curve. Rows progress from intact input through intermediate masking to the blank control. Left panels show five binarized examples and their mean visible foreground-pixel count. The ANN and PING panels show confusion matrices with true digit on the vertical axis and predicted digit on the horizontal axis; each row is normalized to show the distribution of predictions for one true class. The lowest-retention rows reveal collapse toward model-specific default classes rather than structured digit confusions.

Binarization maps every nonzero antialiased MNIST pixel, including intermediate-intensity pixels introduced to smooth digit edges, to full intensity, so spatial retention is not a second measurement of grayscale contrast. Nevertheless, expected input-event count gives a useful first-order bridge. For an otherwise identical binary image encoded at the masking experiment’s 25 Hz ceiling, retaining fraction q gives the same expected event count as retaining the full foreground and using

$$r_{\text{equiv}} = q \cdot 25 \text{ Hz}. \quad (9)$$

Here r_{equiv} is the full-foreground Poisson rate with the same expected event count, meaning the expected total number of input spikes across the image and presentation; q is foreground retention; and 25 Hz is the masking experiment’s reference encoding rate.

Thus $q = 0.1$ maps to $r_{\text{equiv}} = 2.5$ Hz, within the 2–3 Hz transition of the fixed-duration rate curve. At that retention, PING reaches 45%, compared with 49.7% at 2 Hz in the grayscale rate sweep. This numerical alignment supports including a rate near 2.5 Hz in variable-rate training, but it is not an equivalence of corruptions: spatial masking permanently removes locations, whereas lowering Poisson rate preserves all locations in expectation and changes temporal sampling noise.

Appendix: Neo-proposal for filter-matched variable-rate calibration

Neo-proposal

1. **Ask one operational question.** At fixed 200 ms presentation, what is the lowest Poisson encoding rate at which a filter-matched pixel representation still supports useful digit classification? The answer will be used to set the lower endpoint of a subsequent variable-rate PING training run.
2. **Feature generation.** Give every image pixel its own AMPA synapse and uncoupled non-spiking excitatory membrane. Raw spike counts ignore when events arrive and how synaptic and membrane decay weight those events within a finite presentation. The time-averaged voltage instead provides one temporally filtered feature per pixel.

This is a filter-matched pixel representation, not an exact reconstruction of PING’s hidden state. PING combines weighted pixel conductances before membrane integration, whereas this probe integrates each pixel separately and combines the resulting features inside the ANN. For pixel i , draw

$$S_i(t) \sim \text{Bernoulli}(r\Delta t x_i). \quad (10)$$

Update the pixel probe’s excitatory conductance according to

$$g_i^{\text{pix}}(t) = \beta_{\text{AMPA}} g_i^{\text{pix}}(t-1) + w_{\text{probe}} S_i(t). \quad (11)$$

The AMPA decay factor is

$$\beta_{\text{AMPA}} = \exp\left(-\frac{\Delta t}{\tau_{\text{AMPA}}}\right). \quad (12)$$

Each pixel membrane follows

$$C_E \frac{dv_i}{dt} = g_{L,E}(E_L - v_i) + g_i^{\text{pix}}(t)(E_e - v_i). \quad (13)$$

Its instantaneous effective time constant is

$$\tau_{\text{eff},i}(t) = \frac{C_E}{g_{L,E} + g_i^{\text{pix}}(t)}. \quad (14)$$

Here $S_i(t)$ is the input spike from pixel i at timestep t ; r is the maximum encoding rate in spikes per second; Δt is the simulation timestep in seconds; x_i is grayscale pixel intensity; g_i^{pix} is the conductance of pixel probe i ; w_{probe} is a fixed conductance increment shared by all pixels and rates; β_{AMPA} is the conductance retained for one timestep; τ_{AMPA} is the AMPA time constant; C_E is excitatory-cell capacitance; v_i is pixel-probe voltage; $g_{L,E}$ is leak

conductance; E_L and E_e are the leak and excitatory reversal potentials; and $\tau_{\text{eff},i}$ is the conductance-dependent membrane time constant.

Set w_{probe} to $1.2 \mu\text{S}$ for the primary analysis. This is the mean of the configured pre-training PING input-weight distribution, so it preserves the model’s synaptic conductance scale without importing any trained weight or spatial projection. It is not a neutral normalization. The excitatory leak conductance is $0.05 \mu\text{S}$, so one event initially contributes 24 times the resting leak and temporarily shortens the effective membrane time constant from 20 ms to approximately 0.8 ms. The probe therefore measures decodability at a specified conductance operating point, not an amplitude-independent property of the Poisson encoder. Equation 14 is why the ANN cannot simply absorb a different w_{probe} into its learned weights.

Test this dependence by repeating the complete decoder training and held-out rate sweep with w_{probe} set to 0.6 and $2.4 \mu\text{S}$. These half-scale and double-scale controls preserve the model form while testing whether the inferred lower rate is driven by the selected synaptic amplitude. If the practical training floor moves by more than one adjacent rate-grid point, report an operating-point-dependent range rather than a single universal floor.

Restart every pixel probe from the same initial state and retain exactly the first 200 ms. Reduce each presentation to one static feature vector by computing, for every pixel probe,

$$z_i = \frac{1}{T} \int_0^T (v_i(t) - E_L) dt. \quad (15)$$

Here z_i is the time-averaged baseline-subtracted voltage and T is presentation duration. The feature is not divided by encoding rate because PING is not given that rate and experiences its rate-dependent change in mean drive.

3. **Build the variable-rate feature dataset.** Apply the feature-generation recipe in Step 2 to the established MNIST training and held-out partitions, using independent Poisson realizations from seeds 42, 43, 44. Evaluate 0.25, 0.5, 0.75, 1, 1.5, 2, 2.5, 3, 4, 5, 10, and 25 Hz at fixed 200 ms presentation. The grid ends at 25 Hz because the existing PING results establish it as a high-accuracy trained endpoint; the denser low-rate samples are intended to locate the lower decodability boundary. During decoder training, select one listed rate point with equal probability for each image presentation and generate a fresh Poisson realization. Split the training images into decoder-training and validation subsets before selecting regularization or stopping epochs; leave held-out test images untouched.
4. **Train only two mixed-rate decoders initially.** The primary decoder is a conventional ANN with one rectified-linear hidden layer of 1024 units and 10 outputs. It learns its own input projection from the pixel features and its own output weights across all training rates. Train a regularized linear softmax decoder on the same features as a diagnostic of linear accessibility. Neither decoder receives the numerical rate or shares learned weights with PING.
5. **Generate the psychometric curves by inference only.** Freeze both decoders. At every rate, evaluate the same held-out images using fixed Poisson draws shared across compatible decoder and PING seeds, with additional draws used to measure sampling variation. The primary curve is

$$A_r(r) = P(\text{correct} \mid r, \text{mixed-rate nonlinear decoder on Equation 15}). \quad (16)$$

Here A_r is held-out nonlinear-decoder accuracy and P denotes probability. Plot the linear curve beside it, but do not use the linear decoder to set the training range.

6. Report a decoder-relative decodability edge and a practical training floor.

Bootstrap held-out images, Poisson draws, and model seeds to obtain a lower confidence bound L_r at every tested rate. Define the decodability edge as

$$r_{\text{decode}} = \text{lowest } r \in \mathcal{R} \text{ satisfying } L_{r(r)} > \frac{1}{N_{\text{class}}}. \quad (17)$$

Define the practical training floor as

$$r_{\text{train}} = \text{lowest } r \in \mathcal{R} \text{ satisfying } L_{r(r)} \geq a_{\text{use}}. \quad (18)$$

Here the calligraphic $\mathcal{R}$ is the tested rate grid; N_{class} is the number of digit classes; r_{decode} is the lowest rate at which the primary nonlinear decoder is reliably above chance; a_{use} is a predeclared useful-accuracy target; and r_{train} is the lower endpoint used for training. Use 50% correct as the provisional a_{use} value and lock it before inspecting the curve. Report r_{decode} descriptively, but use r_{train} for the training decision. Report the half-scale and double-scale sensitivity results from Step 2 beside the primary estimates.

7. Compare with the frozen PING curve. Compare the nonlinear ANN curve with the fixed-duration PING rate curve in Figure 2. If both fail, neither tested system extracts usable digit evidence at that rate. If the ANN succeeds but PING fails, the temporally filtered pixel representation retains decodable evidence that the constant-rate-trained PING network does not exploit. That gap may arise from PING’s trained input projection, recurrent dynamics, or readout. If PING succeeds where the ANN fails, the pixel-probe summary or decoder is inadequate.

8. Keep the existing spatial results as sanity checks. Reuse the foreground-retention curve in Figure 3 and the equal-event bridge in Equation 9 without rerunning them. Equal-event mapping compares expected input-spike count, not class information. Because Figure 3 uses binarized images while the rate curve uses grayscale inputs, treat the simple mapping in Equation 9 as an order-of-magnitude comparison rather than an exact equivalence.

9. Escalate only if the primary comparison demands it. If the primary ANN succeeds where PING fails, add a decoder that uses the frozen PING input projection to test whether the old spatial weights explain the gap. Add a coarse temporal decoder only if PING succeeds where the time-averaged ANN fails. Train rate-specific decoders only if the mixed-rate decoder shows an unexplained failure, initially restricting them to rates around that transition. These are debugging controls, not prerequisites for choosing the first variable-rate training range.

Relation to prior work

- Wolff and Lindner show that transient voltage statistics under filtered conductance shot noise can depart from a fixed-effective-time-constant approximation, especially at low rates [1]. This supports using the exact finite-window conductance probe in Step 2.

- Brigham and Destexhe show that nonstationary filtered shot noise can produce voltage distributions whose skew is missed by a Gaussian approximation [2]. This supports fresh Poisson simulation rather than synthetic additive ANN noise.
- Quian Quiroga and Panzeri emphasize held-out decoder evaluation and warn that failure of one decoder does not prove that a neural response contains no information [3]. This is why the above-chance decodability edge is described as decoder-relative rather than as an absolute information boundary.
- Warland et al. compare linear and nonlinear population decoders [4]. This motivates the one linear diagnostic beside the primary nonlinear ANN.

Why Bode analysis is deferred

A Bode analysis would describe the temporal bandwidth of a locally linearized synapse-plus-membrane cascade. It would not directly determine whether digit identity remains classifiable or which rate should bound variable-rate training. The exact finite-window probe already answers the calibration question without relying on a linear approximation.

In addition, PING’s effective membrane time constant changes with conductance, so there is no single universal transfer curve. A defensible Bode result would require a family of operating-point-dependent curves and exact gain-and-phase validation. That remains useful mechanistic and thesis material, but it is a separate follow-up rather than a prerequisite for the neo-proposal.

TODO

1. [] Implement and unit-test the finite-window per-pixel synapse and non-spiking membrane probe in Step 2 without changing the production PING engine.
2. [] Generate time-averaged filter-matched features over the fixed rate grid using Poisson seeds 42, 43, 44.
3. [] Train the mixed-rate nonlinear ANN with its own input and output weights, together with the linear softmax diagnostic.
4. [] Freeze both decoders and run the shared held-out rate sweep.
5. [] Repeat decoder training and held-out evaluation at the half-scale and double-scale probe conductances defined in Step 2.
6. [] Bootstrap both curves and report r_{decode} , r_{train} , and their sensitivity to probe conductance.
7. [] Compare the primary ANN curve with frozen PING Figure 2, Figure 3, and the equal-event bridge in Equation 9.
8. [] Select the first variable-rate PING training range using r_{train} as its lower endpoint.
9. [] Run frozen-projection, temporal, per-rate, or Bode follow-ups only if the primary result triggers the conditions in Step 9.

References

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4. Warland, Reinagel & Meister — *Decoding Visual Information From a Population of Retinal Ganglion Cells*. Journal of Neurophysiology, 1997. doi:10.1152/jn.1997.78.5.2336